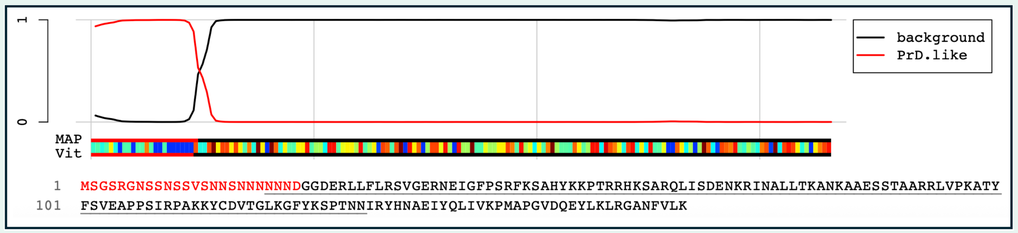
Prions at the End of Their Rope: Telomeres, Ies6, and Some SIRious Business!

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Introduction

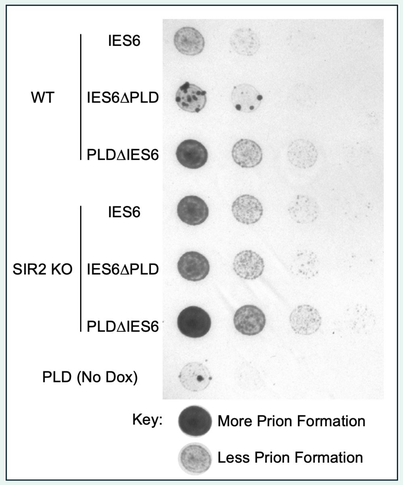
Prion-Like Domain in INO80 Subunit Ies6

The INO80 chromatin remodeling complex regulates transcription, DNA replication, and repair by repositioning, ejecting, or restructuring nucleosomes to modulate DNA accessibility.[1] One of its essential subunits, Ies6, contains a prion-like domain (PLD) – regions of low-complexity amino acid sequences that can drive aggregation and potentially prion formation.



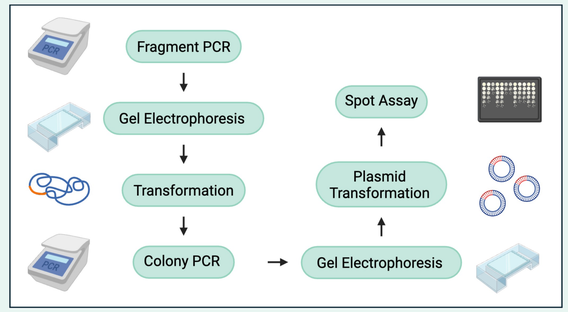
Prions are proteins that adopt alternative conformations capable of self-propagation.[2] Prions such as [PSI+] (from Sup35) or [URE3] (from Ure2) act as epigenetic elements, altering cellular phenotypes without changes in DNA sequence, and once formed, prions are heritable through cell division, as they can 'seed' the misfolding of new proteins.[2]

Telomere-Interacting Proteins as Potential Modulators



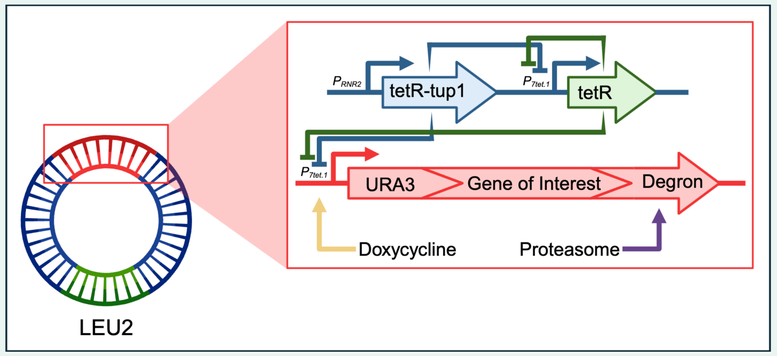
Interestingly, proteins that interact with telomeres – such as sirtuin family (Sir1-4), checkpoint kinases (Tel1), transcription factors (Rap1) – have also been implicated in chromatin remodeling and epigenetic inheritance.[3] These factors regulate telomere silencing, DNA end protection, and checkpoint signaling, but it has been suggested they may also influence prion formation. For instance, Sir2 has been shown to affect prion propagation, raising the possibility that other telomere-interacting proteins could modulate the prion-like behavior of chromatin-associated factors such as Ies6.

Workflow for a Single Spot Assay



The spot assay is a functional readout that measures yeast growth across serial dilutions under different genetic or environmental conditions.

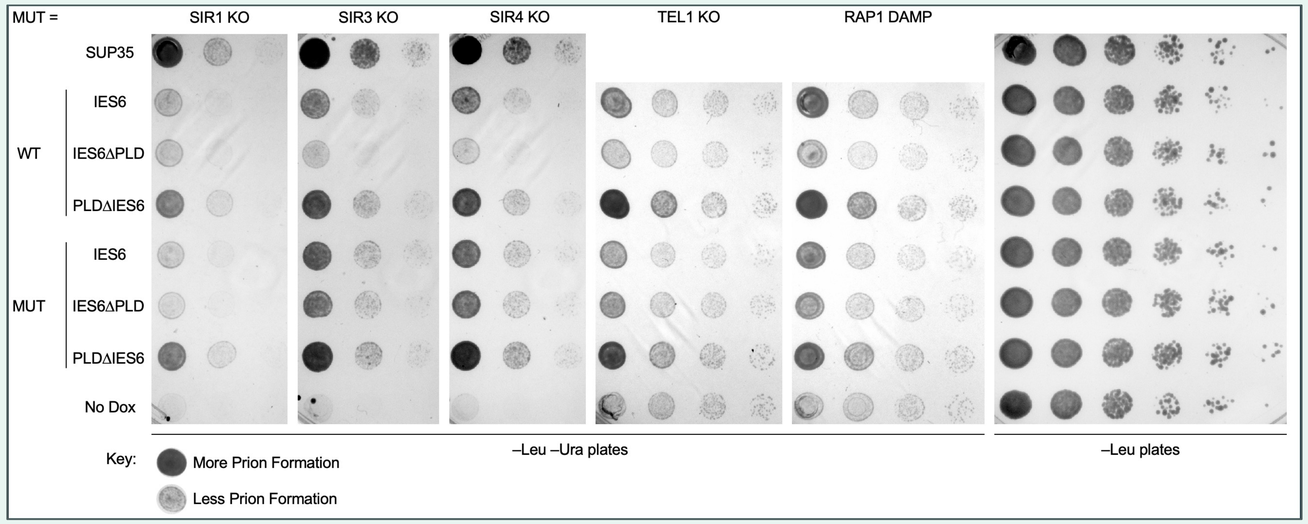
Methods



In the TetR TetR system, a gene of interest (GOI) is placed downstream of a Tet-responsive promoter.[4] In our study, the GOIs are IES6, IES6ΔPLD, and PLDΔIES6. These constructs are transcribed only after addition of doxycycline. Overexpression increases the likelihood of prion formation and protein aggregation, and aggregated proteins attract fewer proteasomes to degrons, leading to reduced degradation. As a result, more prion formation correlates with enhanced cell growth in –Leu –Ura medium, where URA3 expression is essential for survival.

Results

Spot assays were performed on –Leu (control) and –Leu –Ura (reporter) plates using six-fold serial dilutions. For SIR1, TEL1, and RAP1-damp mutants, colony growth on –Leu –Ura was comparable to wild-type (WT), suggesting that these proteins exert little or no effect on Ies6-dependent prion formation under the conditions tested. In contrast, SIR3 and SIR4 mutants displayed stronger growth (darker spots) on –Leu –Ura plates relative to WT. This indicates that loss of SIR3 or SIR4 enhances prion activity, thereby stabilizing URA3 and promoting cell growth in selective medium.



Conclusions

- **No detectable role for SIR1, TEL1, RAP1-damp in this assay.** Spot intensities for SIR1, TEL1, and RAP1-damp on –Leu –Ura are comparable to WT, meaning they have little or no effect on prion formation/aggregation-driven URA3 stabilization under our conditions.
- **SIR3 and SIR4 constrain prion formation.** SIR3Δ and SIR4Δ show stronger growth (darker spots) on –Leu –Ura, indicating more prion formation/protein aggregation, which (by our model) reduces proteasome engagement with degrons, stabilizes URA3, and boosts growth on –Ura. Thus, SIR3/SIR4 act as negative regulators of the prion-dependent signal.
- Together, these results highlight a potential functional link between telomere-interacting proteins and prion-like activity, with SIR2/SIR3/SIR4 acting as key modulators.

Future Directions

- SIR2, SIR3 and SIR4 seem to negatively regulate Ies6 prion formation, but the underlying mechanism remains unresolved. Future work will test whether their effects arise through chromatin silencing at telomeres, direct influence on Ies6 aggregation, or altered proteasome recruitment to degrons. The surprising orientation independence of IES6ΔPLD in SIR3Δ and SIR4Δ strains raises the possibility that loss of these factors causes a generalized increase in protein aggregation, which we will evaluate using additional PLD-fusion constructs.
- To broaden our understanding, we will expand the screen to other telomere-interacting proteins such as Rif1, Rif2, and Ku70/80, and assess prion formation under stress conditions (heat shock, oxidative stress, nutrient limitation) to determine whether Sir3/Sir4 deletion enhances prion activity universally or only under baseline growth. Finally, given the link between Sir2 and autophagy, we will explore epistasis with autophagy pathway mutants.

References

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